URI SPRECHER

PhD. Student, TAU

PROFILE

My name is Uri and I am currently pursuing a PhD in Biology at Tel Aviv University under the supervision of Prof. Miguel Weil.
My research is focused on rare disorders called GSDs. I utilize cutting-edge microscopy imaging technology, omics and sophisticated statistical analysis tools to model fibroblasts from patients and to evaluate potential drugs for these disorders. Through my research, I have established three effective models of GSDs which have helped me to characterize the biological effects of a drug for APBD, a neurogenerative GSD. This drug is currently being provided as compassionate treatment for APBD patients.
I am currently leading a project to identify these novel drugs for other GSDs patients' fibroblasts using various experimental procedures combined with my expertise in data analysis and statistical modeling. The advancements in the fields of AI and bioinformatics have enabled me to utilize my data analysis and computational skills to efficiently define and bring forward these potential drugs for personalized medicine of patients with GSDs. Moreover, complementing the above-mentioned approach, I contributed to a project in Miguel Weil's laboratory that established a patient derived fibroblast model for Huntington disease.
As a highly motivated and dedicated researcher, I am passionate about exploring the intricacies of my area of study and applying my findings and skills to solve real-world problems.

CONTACTS

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GitHub:
https://github.com/Urisprecher

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uri8sp@gmail.com

Address:
Hoofien 7, Tel-Aviv, Israel

SKILLS

- Languages: Hebrew, English
- Programing Languages: Python, R
- Experience with scientific data analysis, bio-imaging, omics and statistics.
- Experience in fundamental concepts in cellular and molecular biology.

EDUCATION

Tel Aviv University
2021 – present
Direct PhD. Track, "Establishing primary fibroblasts from Glycogen Storage Disorders (GSDs) patients as a predictive or personalized drug evaluation model using high content analysis, omics and computational meta-analysis tools", under the supervision of Prof. Miguel Weil, the Shmunis School of Biomedicine and Cancer Research, the George S. Wise Faculty for Life Sciences, Sagol School of Neurosciences, Tel Aviv University

Tel Aviv University
2024
Participation in FoodTech Nation program

Tel Aviv University
2020 – 2022
Master in biology

Tel Aviv University
2017 – 2020
B. A in biology,

"Sieem" Campus at Tel Aviv University
2012
Physical education and sports studies (as a gym instructor and basketball instructor)

Ironi-Daled High School, Tel Aviv
2009-2011
Full matriculation (Extended professions: sports and chemistry)

WORK EXPERIENCE

Golden Heart Flower– GHF Ltd.
2025-present
Scientific Project Manager

Tel Aviv University
2021-present
Teaching assistant – molecular cell biology lab course, the George S. Wise Faculty for Life Sciences

The Good Chocolate LLC
2015- present
Internet and Social Media Marketing Manager
Scientific Advisor

Independent
2012-2013, 2015 - present
Personal fitness coach and youth basketball coach

Teva Bari
2015 - 2018
Operations and Delivery manager

Independent
2011-2012
Flower Marketer

MILITARY SERVICE

Head of a clinic,7th Brigade, Armored Corps
2014 - 2015
The position required compliance with many tasks in unconventional working conditions, responsibility for equipment and manpower and high-level organizational, command and order capabilities, cooperation with senior ranks.

Squad medic in the 7th Brigade, Armored Corps
2013 - 2014
Squad medic in the 7th Brigade, Armored Corps (senior medical authority in the company, responsibility for the health of the company's soldiers, training of the company's soldiers, management of routine and emergency medical incidents)

Combat medic in the 7th Brigade, Armored Corps
2012 - 2013

PUBLICATIONS

1. O. Kakhlon, H. Vaknin, K. Mishra, J. D'Souza, M. Marisat, **U. Sprecher**, S. Wald-Altman, A. Dukhovny, Y. Raviv, B. Da'adoosh, H. Engel, S. Benhamron, K. Nitzan, S. Sweetat, A. Permyakova, A. Mordechai, H. O. Akman, H. Rosenmann, A. Lossos, J. Tam, B. A. Minassian, M. Weil. Alleviation of a polyglucosan storage disorder by enhancement of autophagic glycogen catabolism. EMBO Mol. Med. 2021 https://www.embopress.org/doi/full/10.15252/emmm.202114554

2. S. Gharaba, O. Paz, L. Feld, A. Anastasia, M. Weinrab, N. Muchtar, A. Baransi, A. Shalem, **U. Sprecher**, L. Wolf, H. Wolfenson, M. Weil. Perturbed actin cap as a new personalized biomarker in primary fibroblasts of Huntington's disease patients. Frontiers in Cell and Developmental Biology. 2023 https://www.frontiersin.org/articles/10.3389/fcell.2023.1013721/full

3. K Mishra, S Sweetat, S Baraghithy, **U Sprecher**, M Marisat, S Bastu, H Glickstein, G Ronzitti, J Tam, H Rosenmann, M Weil, E Malfatti and O Kakhlon. The autophagic activator GHF201 can alleviate pathology in a mouse model and in patient fibroblasts of type III glycogenosis. MDPI, biomolecules. 2024 https://www.mdpi.com/2883160

4. S Gharaba‡, **U Sprecher**‡, A Baransi, N Muchtar and M Weil. Characterization of Fission and Fusion Mitochondrial dynamics in HD fibroblasts according to patient's severity status. Movement Disorders. 2024 https://www.sciencedirect.com/science/article/pii/S0969996124002675?via%3Dihub ‡ Equal first author contribution.

5. **U Sprecher**, J Dsouza, M Marisat, D Barasch, K Mishra, O Kakhlon, J Manor, Y Anikster and M Weil. In depth profiling of dihydrolipoamide dehydrogenase deficiency in primary patients' fibroblasts reveals metabolic reprogramming secondary to mitochondrial dysfunction. Molecular Genetics and Metabolism reports. 2024 https://authors.elsevier.com/sd/article/S2214-4269(24)00125-3

6. A. Cohen, A. Mouzon, **U. Sprecher**, Martin Kupiec and Ronit Weisman. TORC2 Coordinates MBF-Dependent Transcription and Restrains Oxidative Stress Responses During DNA Replication Stress in Fission Yeast. JBC. 2026. https://doi.org/10.1016/j.jbc.2026.111242

7. **U Sprecher**, D'Souza J, Mishra K, Muchtar N, Canella Miliano A, Mithieux G, Rajas F, Avraham S, Castellani M F, Kauffman H, Bergman Y, Linial M, Anikster Y, Kakhlon O, and Weil M. Disease-associated programming of cell memory in glycogen storage disorder type 1a. https://www.biorxiv.org/content/10.1101/2023.02.20.529109v2.full.pdf+html 2024, Frontiers in Developmental and Cell Biology, revision.

8. **U Sprecher**, M Gupta, Muchtar N, Sarji S, Edilbi D, Shefer O, Kakhlon O, and Weil M. The Autophagic Activator GHF-201 Can Alleviate Pathology in a Mouse Model and in Patient Fibroblasts of Pompe disease. 2025, iSceince, revision.

9. **U Sprecher**, Mishra K, D'Souza J, Muchtar N, Shalev O, Eliassaf A, Morshina A, Felig J, Garti S, Garti N, Eftekhari P, Young L, Gentry M, Engelmayer N, Binshtok A, Sweetat S, Rosenman H, Kakhlon O, and Weil M. A novel PPP1CC ligand ameliorates in vitro and in vivo pathology in APBD models. In preparation for submission.

10. N. Muchtar‡, **U. Sprecher**‡, and Weil M. Review-Primary fibroblasts as a personalized cellular model for genetic disorders, In preparation for submission. ± Equal first author contribution.

CONFERNCE AND POSTERS APPEARANCES

1. SSIEM Annual International Symposium 2023, held at the ICC Jerusalem (International Convention Center), Israel on 29 August - 1 September, 2023. **Poster presentation.**
2. Global Young Scientists Summit 2024, held at the University of Singapore, Singapore on January 6-13, 2024. **Poster presentation.**
3. Shmunis School retreat held at the Dan Hotel, Tel Aviv, May 20, 2024. **Short lecture.**
4. Israel Society of metabolic disorders annual summit 2024, held at Elma hotel, Zikhron Ya'akov on September 16-17 2024. **Closing Lecture.**
5. Multiomics in Metabolic Disease, Held at Cambridge, London on June 4-6th 2025. **Short lecture + Poster presentation.**

SCHOLARSHIPS AND AWARDS

1. Shmunis Excellence Scholarship 2021/2022.
2. ISMD Researchers from Academia Grant for the SSIEM 2023 Symposium.
3. The Yoran Institute of Genome Research and Personalized Medicine Scholarship.2023
4. The Excellence award in Teaching, the George S. Wise Faculty for Life Sciences, Tel Aviv University 2023.
5. Booster travel grant from the Vice President's office 2024.
6. The Tel Aviv University Prajs-Drimmer institute for Development of Anti-Degenerative drugs. 2024
7. The Ela Kodesz Institute for Research on Cancer Development and Prevention travel grants.2024
8. Cancer Biology Research Center travel grant, 2025
9. The Dan Moran foundation travel scholarship, 2025.
10. Shmunis Excellence Scholarship, 2024/2025.
11. Welcome Connecting Science Bursary award, 2025.
12. The Smolarz family graduate school excellence in research award, 2025.
13. The Ela Kodesz Institute for Research on Cancer Development and Prevention travel grants.2025.